

RISK

AI-Powered Credit Risk Intelligence Platform

Built with LightGBM • SHAP • Groq API • Streamlit • Docker

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NEOSTATS | AI Engineer Role Assignment

github.com/sksaleenagrace/AI-Powered-Credit-Risk-Intelligence-Platform

Business Problem

- **High Loan Default Rates:** Banks lose billions annually from unpaid loans
- **Slow Credit Decisions:** Manual review takes days — losing customers
- **Black-box ML Models:** Cannot explain decisions to regulators & auditors
- **Inconsistent Risk Assessment:** Manual bias leads to unfair loan rejections
- **No Real-time Insights:** Analysts cannot query data without SQL expertise

Solution: AI-Powered Credit Risk Intelligence Platform

Our Solution — 4 Core Modules

EDA

EDA Dashboard

Explore 307,511
loan records visually

ML

Risk Prediction

ML-powered default
probability scoring

SHAP

SHAP Explainability

Why the model
made each decision

CHAT

Talk-to-Data Chatbot

Ask questions
in plain English

EDA Key Insights

307,511

Loan Records

122

Features

8.07%

Default Rate

91.93%

Non-Default

1 Overall Default Rate: 8.07%: ~1 in 12 applicants default — imbalanced dataset needing special ML handling

2 Age & Default Risk: Young applicants (<30 yrs) default 8.4% more — avg defaulter age is 40.8 yrs

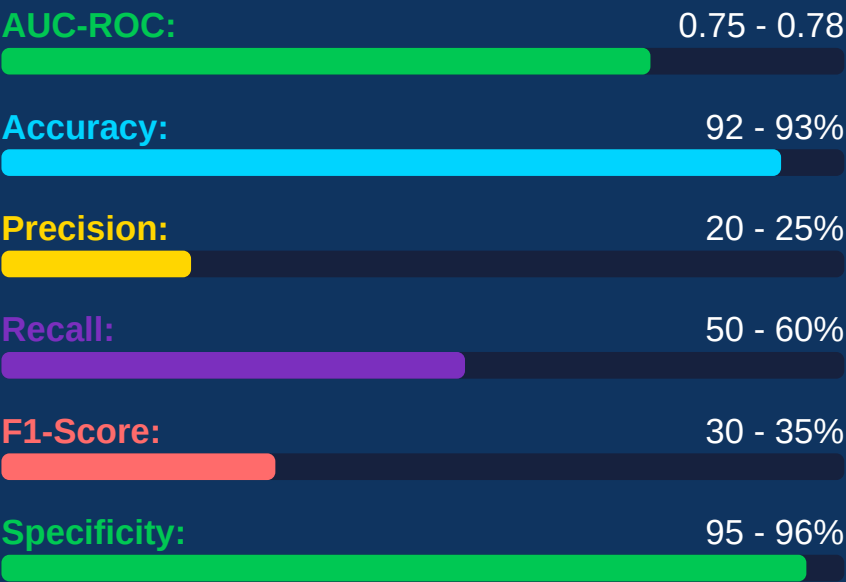
3 Income Impact: Defaulters earn 9.1% less median income (\$135K vs \$148.5K for non-defaulters)

4 Gender Differences: Males default at 10.14% vs females at 7.00% — 44.9% higher male default rate

5 Education Level: Academic degree: 1.83% default vs Lower secondary: 10.93% — 497% difference!

ML Model — LightGBM

Model Performance Metrics



Model Configuration

Algorithm:	LightGBM (Gradient Boosting)
Imbalance:	scale_pos_weight = 11.39
Train Set:	246,008 samples (80%)
Test Set:	61,503 samples (20%)
Features:	134 (122 original + 12 engineered)
Early Stop:	50 rounds patience

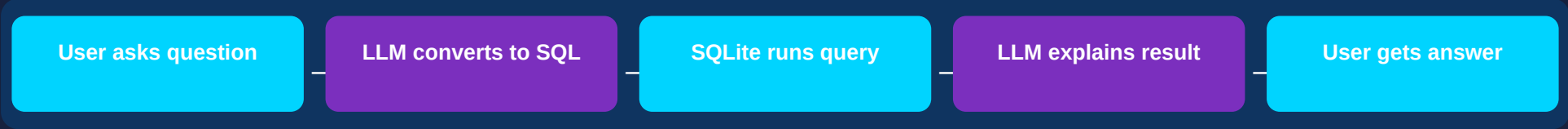
SHAP Explainability

SHAP reveals WHY the model made each decision — satisfying banking audit requirements



Talk-to-Data Chatbot

Powered by Groq API (Llama 3.3-70B-Versatile) — 100% FREE!



Question	SQL Generated	Answer
How many people defaulted?	SELECT COUNT(*) WHERE TARGET=1	24,825 people (8.07%)
Average income of defaulters?	SELECT AVG(AMT_INCOME) WHERE TARGET=1	\$135,000 median
Which gender defaults more?	SELECT CODE_GENDER, AVG(TARGET)...	Males: 10.14%
Average loan amount?	SELECT AVG(AMT_CREDIT)	\$538,396
How many own a car?	SELECT COUNT(*) WHERE FLAG_OWN_CAR=Y	~104,128 applicants

Docker Deployment

Run the ENTIRE platform with ONE command:

```
docker-compose up --build
```

Then open: <http://localhost:8501>

- 1 Clone Repository:** `git clone https://github.com/sksaleenagrace/AI-Powered-Credit-Risk-Intelligence-Platform`
- 2 Configure API Key:** Copy `.env.example` to `.env` and add your FREE Groq API key
- 3 Download Dataset:** Place `application_train.csv` in `data/` folder from Kaggle
- 4 Launch Platform:** `docker-compose up --build` (installs everything automatically)
- 5 Access Platform:** Open <http://localhost:8501> — all 4 modules ready instantly!

Tech Stack

LightGBM

[ML Model]

Gradient boosting for loan default prediction with class imb

SHAP Library

[Explainability]

SHapley Additive exPlanations for model interpretability

Groq API

[LLM]

Llama 3.3-70B-Versatile for natural language to SQL conversi

Streamlit

[UI Framework]

Multi-page interactive web application with dark theme

SQLite

[Database]

Lightweight database for chatbot data storage and queries

Docker + Compose

[Deployment]

Containerized deployment — runs with single command

Matplotlib/Seaborn

[Visualization]

EDA charts, SHAP plots and data visualizations

Python 3.10

[Language]

Core development language with pandas, numpy, scikit-learn

Thank You!

AI-Powered Credit Risk Intelligence Platform

Key Achievements

- AUC-ROC: 0.75-0.78
- Accuracy: 92-93%
- 5 Working NL Queries
- Full Docker Deploy
- SHAP Explainability

Future Improvements

- XGBoost & Neural Networks
- Real-time Prediction API
- User Authentication
- More Chatbot Capabilities
- Ensemble Methods

Repository: github.com/sksaleenagrace/AI-Powered-Credit-Risk-Intelligence-Platform

Assignment: NEOSTATS AI Engineer Role | By: SK Saleena Grace